

***Dictyostelium discoideum* Aggregation Fields as Excitable Media**

A. J. DURSTON

*Department of Biophysics and Theoretical Biology,
University of Chicago, 920 East 58th Street,
Chicago, Illinois 60637, U.S.A.*

(Received 7 May 1973)

This paper introduces some experimental investigations of wave propagation during aggregation of the cellular slime mould *Dictyostelium discoideum* (Dd) (Durstun, unpublished results). It provides a basis for interpretation of the experimental findings by describing the excitable properties of aggregating Dd cells and some expectations about the nature of Dd wave propagation and the properties of the pacemaker regions which organize aggregation centers.

1. Introduction

A central problem in contemporary biology is to account for the emergence of spatiotemporal order in developing organisms. During development of a typical metazoan, an original embryonic structure, which is usually a unicellular fertilized egg, may increase in mass by a factor of 10^{10} and in cell number by a factor of 10^{12} to make an adult containing as many as a hundred cell types arranged in complex and characteristic patterns and shapes. The whole process is reliable and invariable in space and time under a spectrum of environmental conditions. An adequate description of any embryogenesis in molecular terms would appear to be an impossibly complex task.

The problem is simplified if, as a first level of analysis, we attempt to describe embryogenesis in terms of activities of single cells, without necessarily requiring a detailed understanding of such activities. A brief appraisal reveals that a very few categories of activity are involved; namely that cells move, secrete, make and break intercellular contacts, differentiate, divide and grow. Such activities are repeated many times by individual cells and cell lineages and are precisely controlled and co-ordinated in space and time. This formulation (Robertson & Cohen, 1972) leads to the notion that global control systems must exist to co-ordinate these unitary activities of cells and that the key to understanding morphogenesis and pattern formation in embryogenesis will be

to detect and describe such control systems (Robertson & Cohen, 1972; Wolpert, 1969; Goodwin & Cohen, 1969; Cohen, 1971). A number of investigations have now been directed towards this goal, using various experimental systems (Lawrence, Crick & Munro, 1972; Wilcox, Michison & Smith, 1973; Webster, 1966; Cohen & Robertson, 1971*a*). Such systems should be complex enough to pose the germane questions yet very simple, in order to facilitate their analysis.

This paper and a following paper (Durston, unpublished results) report investigations which are a contribution towards understanding morphogenesis in a suitable experimental system. The system is the cellular slime mould *Dictyostelium discoideum* (Dd) and it meets the criteria defined above, since it has a simple, standard morphogenesis and a simple reliably defined pattern of differentiation (Bonner, 1944). It also has the experimental advantages of a micro-organism including a short life cycle, the capacity to grow on axenic media (Schwalb & Roth, 1970) and a convenient genetic system (Loomis, 1969*a*). The investigations are of cell aggregation which is the first stage of Dd morphogenesis. Previous investigations already show that this simple morphogenetic movement is governed by a control system of some complexity. The important findings will be described, following a brief account of the Dd life cycle.

Growing cells of Dd are isotropic amoebae, 5 to 10 μ in diameter, which move freely and independently over any suitable moist surface and feed in their natural state by phagocytosing bacteria (Bonner, 1967). The food bacteria are detected and caught by chemotaxis, the chemotactant being folic acid (Pan, Hall & Bonner, 1972). The amoebae grow and divide by binary fission with a minimum doubling time of 4 hr as long as bacteria or other suitable nutrients are supplied. If the food source becomes exhausted, the amoebae enter upon a multicellular morphogenetic phase which has the following standard time course, under our conditions (Robertson & Cohen, 1972). Initially, the starved cells move randomly, like vegetative cells but they then become quiescent, within about 30 min of removal of the food source. The quiescence lasts for about 8 hr and then the amoebae begin to aggregate. Investigations show that the quiescent phase (interphase) is a period of differentiation, during which a number of competences needed for aggregation are gained (Cohen & Robertson, 1971*b*) and during which specific biochemical changes occur in the cells (Loomis, 1969*b*). At the onset of aggregation, Dd amoebae commence polar movements towards certain individuals which initiate aggregation centres. Pseudopodia are now extruded from one end of the anisotropic cell only, in contrast to the situation in the isotropic vegetative amoebae, where a randomly oriented succession of pseudopods is extruded (Shaffer, 1962, 1964). As aggregation proceeds, the

initial pattern of aggregates changes and new centres emerge and old ones disappear until finally a stable pattern of aggregates is formed, each containing 10 to 10^5 cells, the number depending on the initial cell density (Bonner & Dodd, 1962). As Dd aggregates develop they display a characteristic morphology; the aggregating cells form streams, radiating from the centre, in which specific contacts are made between each cell and the back of the preceding cell (Gerish, 1968). The "strings" of cells thus formed persist for some time during later Dd morphogenesis and the intercellular contacts may have a special role in the multicellular organization of Dd as discussed in section 3 (p. 489) (Gerish, 1968). The formation of streams is a consequence of certain characteristics of the Dd aggregation control system (Cohen & Robertson, 1971*b*) and section 7 (p. 502). The late Dd aggregate, which is an approximately hemispherical mound of cells, gives rise to an approximately cylindrical column of cells, termed the grex or slug stage. The slug erects vertically from the central part of the aggregate, topples over and then usually migrates for some time. Following a period of migration, determined by environmental conditions, the slug rounds up and commences to form a fruiting body. A tapering stalk, bounded by an acellular cellulosic sheath and containing vacuolated thin walled stalk cells is extruded downwards into the contracted slug and elongated by addition to the sheath and its contents from the top end, so that the mass of undifferentiated cells is raised from the substrate (Bonner, 1944). The extension of the stalk continues for a defined period, and the remaining cells then undergo a differentiation to become mature spores; they gain a rigid cell wall and become ellipsoidal in shape (Bonner, 1944). The mature stalk cells are dead (Raper & Fennell, 1952) and the spores, if deposited in a moist environment (Ceccarini, 1966; Cotter & Raper, 1968) hatch to release single amoebae which may either feed or aggregate directly, depending on the presence or absence of a food source. All of the later stages of Dd development are characterized by the possession of a morphologically distinct tip. This is a nipple-shaped association of cells which appears first in the apex of the late aggregate, continues as the anterior portion of the slug and is the apical portion of the nascent fruiting body. The tip has been shown to be an organizer (Raper, 1940; Farnsworth, 1972) in the same sense as the dorsal lip of the blastopore in the amphibian embryo and the hypostome in hydra are organizers (Spemann & Mangold, 1924; Webster & Wolpert, 1972) and it controls the movements of cells in late aggregates, migration of the slug and erection of the fruiting body. Paper 2 in this series (Durstun, unpublished results) contains some results concerning the pacemaker properties of the tip which are of interest for interpreting its function as an organizer.

The aggregation stage of Dd development is particularly suitable for analysis

because it occurs via movements of a two-dimensional array of separate cells. These movements can be monitored by time lapse cine-micrography, and if measured in suitable experimental paradigms can yield cellular and global parameters relating to the control of aggregation. This feature has led to a series of investigations directed towards elucidating the Dd aggregation control system (Cohen & Robertson, 1971*a,b*; Robertson, Drage & Cohen, 1972). The approach has been to make a theoretical model which realistically accounts for the known global features of Dd aggregation in terms of known or projected cellular activities. The model is used as a frame of reference for experimental investigation by using it to make qualitative and quantitative predictions about features of aggregation and relationships between global and cellular parameters. The predictions are tested and further investigations based on the predictions are made, and the model is modified or extended appropriately. A full description of these investigations cannot be made here; what will be given is an account of some experimental facts which led to the original model, a description of the model and an account of some predictions which have been tested.

Time lapse films of Dd aggregation show that the movements of aggregating cells are globally organized. Individual cells move towards aggregation centers in a periodic pulsatile fashion and movements of cells responding to one aggregation center are co-ordinated, so that periodic waves of movement pulses propagate out from the center (Gerisch, 1968; Shaffer, 1957, 1962). The field of cells controlled by one center may be centimetres in diameter and may contain more than 10^5 cells. The aggregation waves propagate at a constant velocity of about $45 \mu/\text{min}$ (Gerisch, 1968) and when waves from neighboring centers collide or when waves meet a boundary to the field, the waves are extinguished. Thus, a field of Dd cells appears to be a classical example of an excitable medium and will have properties in common with certain electrical circuits (Brunisard, 1960) excitable chemical reaction mixtures (Zaikin & Zhabotinski, 1970) and heart and nervous tissues (Wiener & Rosenbluth, 1946; Katz, 1966) in metazoa. It may be deduced that Dd aggregation involves the relaying of signals by cells in aggregation fields (Shaffer, 1957). Some additional findings help to clarify this picture. These are that the aggregative movements of Dd cells are chemotactic (Bonner, 1947), and that the chemo-tactant is probably 3'5' cyclic AMP (cAMP) (Bonner *et al.*, 1969; Konijn, Barkley, Chang & Bonner, 1968). Dd cells secrete and will respond chemo-tactically to cAMP during their aggregative stage but not during their vegetative stage (Bonner *et al.*, 1969) and aggregating Dd cells also secrete a cAMP phosphodiesterase (PDE) (Chang, 1968).

Some features of the theoretical model to account for these observations are as follows. Dd cells in centers are assumed to secrete periodic pulses of

cAMP. When cells within range of these diffusing signals sense a super-threshold level of cAMP, they respond in a dual fashion; they secrete an equivalent pulse of cAMP, thus relaying the signal, and they make a movement step towards the signalling cell. The movement step for a wild type Dd cell covers about $20\ \mu$ and lasts about 100 sec (Cohen & Robertson, 1971a), and approximate calculated values for the cAMP pulse amplitude from one cell and the threshold concentration of cAMP for relaying are of order 10^{10} molecules and 10^{-6} M, respectively (Drage, Robertson & Cohen, unpublished results). A lower concentration of cAMP ($\sim 10^{-8}$ M) will elicit a chemotactic response but not a relaying response (Drage, Robertson & Cohen, unpublished results). Estimates of the extracellular activity of PDE suggest that aggregation signals are impulsive, lasting about 0.1 to 0.2 sec above the threshold at the site of their production (Cohen & Robertson, 1971a). After responding to a signal, each cell enters a refractory period during which the cell cannot make a further signal or a movement step in response to another signal (Cohen & Robertson, 1971a). There may be two refractory periods, one for the relaying response and one for the chemotactic response. The refractory period for relaying is an age dependent variable which varies between 120 and 420 sec (Durstun, unpublished results; Drage, Robertson & Cohen, unpublished results). The movement refractory period has not been determined. Relaying refractivity ensures unidirectional wave propagation through aggregation fields and results in the extinction of waves which meet other waves or non-excitable barriers. After the end of the refractory period, each cell again becomes sensitive to stimulation and fields of cells can therefore propagate appropriately timed periodic signals.

The model predicts that aggregation waves may be initiated by periodic local application of pulses of cAMP to fields of aggregation competent Dd cells. Pre-aggregative fields of Dd cells have been stimulated with locally applied pulses of cAMP, of suitable amplitude generated electrophoretically from a cAMP-microelectrode (pulser) (Robertson, Drage & Cohen, 1972). The periodically applied pulses initiated periodic waves of cell movements, and the electrode effectively simulated an aggregation centre. Thus, this prediction is confirmed. Calibrated pulsers are now being used to obtain experimental values for thresholds for the relaying and movement responses, to be compared with those calculated from other data (Drage, Robertson & Cohen, unpublished results).

Paper 2 in this series (Durstun, unpublished results) reports detailed measurements and observations of two global aspects of Dd aggregation. These are the geometries of Dd aggregation waves and the periodicity of propagation of waves from aggregation centers. The measurements are

interpreted in terms of observed and projected cellular parameters to provide the basis of a model for pacemaker activity in Dd aggregation.

The interpretations make use of established concepts concerning wave propagation in excitable media, and the remainder of this paper provides a basis for the interpretations by describing the excitable properties of Dd cells and some consequences for the morphogenesis of Dd aggregates.

2. Excitable Media

Fields of aggregating Dd cells are excitable media. They therefore share certain generic properties with nervous and muscular tissues (Wiener & Rosenbluth, 1946; Katz, 1966) excitable chemical reagents (Zaikin & Zhabotinski, 1970; Winfree, 1972) and chains of multivibrators (Brunisard, 1960). These properties are as follows.

Excitable media propagate waves of excitation which do not damp or gain in amplitude and which travel at a velocity which is dependent on the properties of the medium and constant for homogeneous media. These waves arise because the elements of the medium in the wavefront generate unitary signals which excite neighbouring elements ahead of the wavefront to produce identical signals. For a medium such as a Dd aggregation field, the velocity of propagation of the excitation depends upon the delay between reception of a signal and signal generation in the units of the medium, the range of the excitatory signal and the rate at which it is transmitted between units of the medium. Experimental findings show that the wave propagation velocity in fields of Dd cells is approximately constant at 45 μ /min (Gerisch, 1968).

Waves of excitation propagating at constant velocity through a uniform medium will propagate for infinite distances and will retain a predictable geometry provided that they do not meet other waves or boundaries to the medium and provided that the medium is homogeneous and isotropic.

Excitable media show refractivity: the elements of the excitable medium enter a defined refractory period following excitation during which they cannot propagate a successive signal. Consequently, waves of excitation are succeeded by refractory zones of definite width, the width being determined by the duration of the refractory period and the velocity of wave propagation. The refractory properties of excitable media guarantee that wave propagation is unidirectional and that waves meeting other waves or boundaries to the medium are extinguished. In Dd, the refractory period is a time-dependent variable, varying between > 6 and < 3 min (Drage, Robertson & Cohen, unpublished results). This property will be important for interpreting the results obtained in experimental investigations of the period of wave propagation for Dd aggregation centers (Durstun, unpublished results).

In most excitable media, including fields of slime mould cells, the duration of the refractory period is finite and medium regains its excitable properties upon leaving its refractory phase. Such excitable media are termed regenerative and will propagate waves of excitation periodically if stimulated with an appropriate periodic signal.

3. Dd Aggregation Fields as Excitable Media

Dd aggregation fields have some specific properties which define the nature of their activity as excitable media. Three properties will be described at this point. A fourth property of importance (the movement response of cells to an excitatory signal) is discussed in a separate section (section 7, p. 502). The first property is the polarity of wave propagation in Dd aggregation fields. In early aggregation fields the excitable units are separate cells which communicate via freely diffusing signals (pulses of cAMP). The signals have a range ($60\ \mu$) which exceeds the diameter of the cell ($10\ \mu$) (Cohen & Robertson, 1971a) and diffuse symmetrically through the extracellular medium. Consequently, early Dd aggregation fields are approximately isotropic excitable media, and will propagate waves with any orientation as has been observed experimentally (Durston, unpublished results). During late aggregation, Dd cells enter streams which radiate from aggregation centers. Cells in streams are joined by special head to tail junctions the front of each cell being oriented towards the aggregation center and joined to the back of the preceding cell (Gerisch, 1968). Cells approaching streams orient towards the head to tail junctions between cells or else towards the tail end of a stream before they enter the stream (Shaffer, 1962). It is very likely that intercellular chemotactic signals now pass only through these special intercellular junctions without much leakage into the extracellular medium and that cells in streams are polar with a front, signal sensitive end and a back, signal releasing end. The existence of cellular polarity is also indicated by observations of the chemotactic movements of single cells (see section 7, p. 502). Cell communication in late Dd aggregates may thus bear resemblance to synaptic communication between neurons (Katz, 1966). The above observations suggest the possibility that aggregation streams should be locally anisotropic excitable media. Since all cells in streams point towards the aggregation center and cells are joined by junctions which may rectify signals, signal propagation may be possible away from but not towards aggregation centers. Some early observations by Bonner (1950), seem to indicate that this prediction is true. Bonner excised sections of aggregation streams and replaced them with reversed polarity. His observations imply that reversed stream sections lost the capacity to propagate waves of excitation

away from the aggregation center.†

The second property is that Dd aggregation fields are discrete excitable media. The elements of the medium are separate excitable entities (cells) which are excited or not according to intercellular interactions. On the simplest model (that described in the Introduction) (Cohen & Robertson, 1971*a*), the only relevant interaction is an excitatory signal, namely a pulse of cAMP which induces secretion of an identical signal, followed by a refractory period in the responsive cell. The alternative case to a discrete excitable medium is a continuous medium in which excitable elements are not distinguishable at the level of observation, e.g. are molecular species. In such media, interactions between zones of different phase during wave propagation may be very complex and may involve diffusion of a number of reactants. An example of a continuous excitable medium is the Zhabotinski-Zaikin reagent (Zaikin & Zhabotinski, 1970; Winfree, 1972). This is an appropriate solution of malonic acid, bromate ions, cerous ions and sulfuric acid and a suitable redox indicator (e.g. ferroin), which may propagate visible waves of excitation. The waves can occur in an approximately two-dimensional form if the solution is dispensed in a thin layer in a petri dish and they bear striking resemblances to slime mould aggregation waves.

The third property of importance concerns the capacity of Dd cells to make autonomous signals. Dd cells spontaneously aggregate at eight hours after starvation at 22°C. At this time all cells can make chemotactic movements (since all cells enter aggregates). It is also probable that the majority of cells relay chemotactic signals (i.e. secrete signals in response to external signals). This conclusion is indicated because Dd cells now enter aggregation streams and streams can be deduced to contain a substantial proportion of cells which relay signals (Cohen & Robertson, 1971*b*). Aggregation is initiated by the genesis of autonomous periodic signals by entities which are likely to be

† Bonner's diagrams show that when a 200 μ section of an aggregation stream is removed and replaced with reverse polarity, it maintains contact with the proximal part of the original stream (and thus with the center) but rounds up and thus separates from the distal part of the stream. The polar cells in the stream meanwhile round up, then become polar again; the resumption of polarity proceeding proximodistally. He also shows that cells in streams may respond to a center which has been placed distally to them, by separating from the stream in a disto-proximal order, making U-turns (or else rounding up and repolarizing) and moving to the distal center.

I interpret these observations to mean that Dd cells in streams have a polar movement response and so cannot move to a distal signal source unless they break their cellular contacts and turn around or else repolarize, and also that Dd cells in streams have a polar signal relaying response such that they cannot relay signals unless they are moving (and possibly in contact) with correct polarity. This second postulate is to account for the observations that centers fail to attract distal sections of streams via reversed stream sections and that repolarization of cells in a stream section occurs in a proximodistal wave. The wave takes about 16 min to propagate over a section which is 200 μ in length, i.e. it is 4 times slower than the normal propagating signal.

single cells (Durstun, unpublished results). It is of interest to know how many potentially autonomous entities exist in aggregating Dd cell populations, that is to say, if they are cells, the probability of autonomous competence in a cell. The incidence of autonomous competence has been measured by scoring incidence of aggregation in small populations of cells (Konijn & Raper, 1961). The measurements show that within their period of observation populations of about 100 cells aggregate with 50 to 90% probability. The measurements indicate that only about one autonomous cell (or other autonomous entity) arises per 100 cells, i.e. that potential autonomy is rare among Dd cells. This characteristic may contribute to emergence of stable natural pacemakers during Dd aggregation, since a large proportion of the autonomous entities will act as pacemakers. The expression of autonomy among aggregating Dd cells contrasts with the situation in another excitable medium; namely, the smooth muscle of the canine small intestine (Code & Szurszewski, 1970). The intact intestine spontaneously exhibits spatiotemporal waves of electrical activity. The waves are initiated at the proximal end of the intestine and propagate distally. If the intestine is transected and the parts are reunited, waves propagate both in the proximal and in the distal parts but propagate with a decreased frequency in the distal part. These observations indicate that all parts of the canine small intestine can autonomously initiate waves but that the proximal end of the intestine naturally acts as a pacemaker because it has a faster autonomous frequency. The autonomous competence of the distal intestine is usually suppressed. Transection breaks electrical contact between the proximal and distal parts of the intestinal muscle and so permits expression of the autonomous competence of the distal part.

4. Some Geometrical Considerations

Since early Dd aggregation fields are approximately two-dimensional, uniform isotropic excitable media which propagate waves at a constant velocity with no preferred orientation, I make some predications about the geometries of wave-fronts and relationships between wavefront geometry and types of pacemaker region. A point source pacemaker region (for example, an autonomously firing cell) which propagates waves through a uniform isotropic medium should propagate wavefronts which are concentric rings [Fig. 1(a)]. The temporal and spatial separations of successive rings should be maintained if waves propagate at constant velocity; the wavelength will be the product of the temporal period of wave propagation and the wave propagation velocity. Concentric rings are, in fact, observed from early Dd centers, indicating that early aggregation fields are uniform isotropic excitable media and that early pacemaker regions are point sources. It is likely that such early pacemaker

regions consist of single cells and there is evidence that they make periodic signals (Durstun, unpublished results). Three kinds of wave forms other than concentric rings are expected. These are: single and multiple (nested) spirals and pairs of opposed spirals [see Fig. 1(b), (c), (d)]. Single spirals would arise from cooperative pacemakers, in which one wavefront propagates continuously around a closed loop of cells. Multiple spirals would arise where more than one wavefront propagates around the same loop. Pairs of opposed spirals would arise where wavefronts propagate counter-rotationally around two adjacent loops. The spirals would in each case be involutes to the pacemaker loop and Archimedean spirals if the loop had a fixed circumference and there-

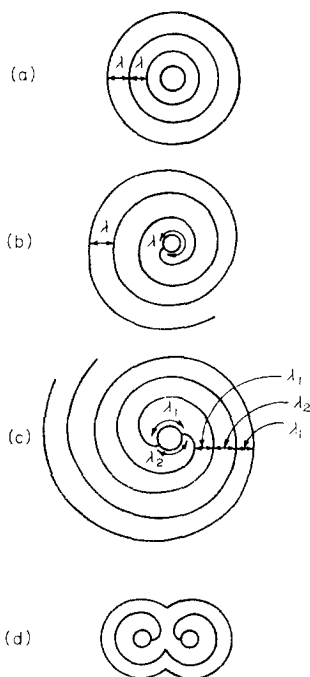


FIG. 1. Wave forms expected in a homogeneous isotropic, excitable medium.

(a) A point source pacemaker would propagate concentric waves. If the source were periodic, the wavelength (λ) would be constant and equal to T (the product of the period of the pacemaker and the velocity of wave propagation).

(b) Spiral wave forms will be involutes to pacemaker loops. For a single spiral, λ will equal the circumference of the loop. If the circumference of the loop is constant an Archimedean spiral will result.

(c) Multiple (nested) spirals will have wavelengths which are equal to the separations of successive waves on the pacemaker loop. If these are unequal on a constant loop, the spiral will propagate beats.

(d) Opposed pairs of spirals may propagate approximately concentric wave forms.

fore a fixed pacemaker period (Winfree, 1972). The wavelength for a single spiral would be equal to the circumference of the effective pacemaker loop; for a multiple (nested) spiral, it would be equal to the separation of successive wavefronts circulating around the loop. For pairs of opposed spirals which are very close together, the propagating wavefronts would appear to be approximately concentric. All pacemaker regions producing spiral wave propagation are expected to arise from interactions between wavefronts and irregularities in the excitable medium. There are good reasons to expect that those classes which have been described will occur; as will be described in section 6, and all of these classes have indeed been seen in aggregation fields (Durstun, 1972; Gerisch, 1968, 1971, and personal communication). Spirals have predictable properties as pacemakers and these will be described in the next section.

5. Periodicity of Wave Propagation from Pacemaker Regions

Excitable slime mould cells have a number of quantifiable parameters which determine the periods with which pacemaker regions propagate waves. Two of the most important of these parameters are the refractory period (T_r) for which a cell is insensitive to further excitation after relaying a signal, and the autonomous period (T_a) of signal generation by an autonomously firing cell or other autonomously periodic pacemaker region.

In spiral pacemaker regions which operate by wave propagation, from a pacemaker loop, pacemaker activity does not involve autonomous firing of pacemaker cells. Each cell in the loop is fired passively by its anterior neighbours and relays the signal to its posterior neighbours. The period of the spiral depends only on the circumference of the pacemaker loop, the number and spacing of any wavefronts propagating around the loop, and the propagation velocity of these waves.

The sizes of pacemaker loops in new born spirals are determined by the ways in which spirals are initiated, as will be described in the next section. The initial spiral loop will be the smallest possible loop in its local area of the aggregation field (minimum loop) if the spiral forms in a continuously excitable aggregation field but it may exceed this minimum size if the spiral forms in a field which contains inexcitable areas (holes). In the latter case, the loop will be the circumference of a hole, and it will evolve to a minimum loop, via cell movements, as described in section 7.

The period of a spiral can therefore either continuously be a minimum determined by the size of a minimum loop pacemaker, or, it can initially be a long period determined by the size of a hole and then evolve to a minimum period.

The size of a minimum spiral loop, the period of a minimum spiral and its stability are properties which will be determined by the excitable properties of Dd cells.

The simplest acceptable model for Dd wave propagation, due to Cohen & Robertson (1971*a*), and outlined in section 1 (p. 486) proposes only one cell interaction. Cells are hypothesized to secrete a pulse of cAMP in response to an extracellular cAMP signal and to enter a defined refractory period after making such a pulse. During its refractory period, a cell cannot make a cAMP pulse in response to a successive signal. According to this model, the minimum pacemaker loop for a spiral would have a path length equal to the refractory wave length (λr) (i.e. the distance between the front of a linear wave and the back of its refractory zone). Loops shorter than λr could not propagate circulating waves, since the wave front would meet cells while these were still in their refractory state. Loops longer than λr would evolve to λr loops via intra loop propagation if they contained excitable cells. All loops would eventually contain excitable cells because of aggregative cell movements (as described in section 7). This formulation predicts that minimum spirals should be stable with a period equal to the refractory period (Tr). In Dd Tr is a monotonically decreasing time dependent variable, which is about 7 min at the beginning of aggregation and about 2 min at the end of aggregation (Drage, Robertson & Cohen, unpublished results). Periods of minimum spirals would therefore be monotonically decreasing with time in this range.

Dd spirals have at least one property which is not accounted for by the simple proposition given above. This is that Dd (Nc-4) spirals are often unstable and typically make a switch from spiral to concentric wave propagation (Durstun, unpublished results). In view of this fact, it is relevant to consider what can happen to minimum spirals if Dd cells have properties other than those described above. This question is complex and cannot be explored in depth at this time, since we need and do not have an adequate parametric description of Dd wave propagation. At this stage, it is appropriate to state some general considerations.

The center of a spiral has unique geometry. In virtually all parts of an excitable medium other than the middle of a spiral, the front of a wave of excitation and the back of its refractory zone are separated by λr . In the center of a spiral, a wavelength is disposed on a part or the whole of the circumference of a central core and the spatial relations of the wave are therefore changed. For example, in a λr spiral, the whole circumference of the central core is one wavelength. In such a spiral, two cell states which are separated by $\lambda r/2$ during normal (linear) wave propagation would be separated by $\lambda r/2\pi$ across the core if the core were circular.

The unique geometry of spirals could lead to local modification or preven-

tion of wave propagation in the spiral core. No explicit statement of the possibilities can now be made, but it is clear that any modification, if it exists, would arise due to short range cell interactions. These would not prevent or affect linear wave propagation because of the spatial relations between phases in a normal wave. If the outcome of the local effect on wave propagation were, for example, that minimum spirals have a core, exceeding λr in circumference which is an appropriate continuous source of cAMP, then minimum spirals would switch to quasi concentric wave propagation at the refractory period (Cohen, personal communication).†

In fields controlled by a periodic autonomous cell, the periodicity of wave propagation (T) will depend on the refractory period of the field (Tr), the autonomous period of the cell (Ta) and possibly on a third parameter; the duration of the signal (τ). If Tr exceeds Ta , then waves will be propagated at the first multiple of Ta which exceeds Tr . ($T = mTa$; $mTa > Tr > m-1(Ta)$). The above statements are true for the case of an excitable medium in which the duration of the signal (τ) is very short compared to the refractory period (Tr) ($\tau \ll Tr$). This is almost certainly the case for wild type Dd ($\tau = 0.1$ to 0.2 sec; $Tr = 180$ to 420 sec) (Cohen & Robertson, 1971a; Drage, Robertson & Cohen, unpublished results) but may not be so for morphogenetic mutants, where factors affecting τ (e.g. PDE level) may be altered. If τ is of significant duration and Tr is sufficiently similar to, but longer than Ta or a multiple of Ta , then rhythmic wavetrains (beats) may be anticipated, of the form where n waves with $T = Ta$ or $T = mTa$ are succeeded by one wave with $T = 2Ta$ or $T = m+1(Ta)$ (Krinskii, 1971).

Considerations of relationships between Ta and Tr but not among Ta , Tr and τ have been important for interpreting the values obtained for T in the results of the investigations described in paper 2 in this series (Durstun, unpublished results).

It is desirable to note that the same considerations as stated above apply if we consider the case of an autonomously firing cell in which the refractivity properties of the cell are the same as those of other cells in the field and Ta is the period of an oscillator in the cell which may entrain or may not entrain firing of the cell according to the phase (sensitive or refractory) of the signal generation cycle. This formulation of the interactions possible between Ta and Tr is a possible alternative to that given above.

A further complication arises, for the consideration of pacemaker activity by autonomous cells, when we consider that more than one autonomous cell

† If the core of a non- Tr spiral secreted a continuously excitatory cAMP signal, the outcome would be spiral propagation around part of the core and concentric propagation from the rest, leading to a "kinky" spiral. The movement response of the cells (tangential in the kink and more or less centripetal elsewhere) would lead to removal of the kink and a transition to a quasi-concentric form as each wave propagated outwards.

may exist in each pacemaker region. If autonomous signal production by these separate cells is entrainable to external signals (as is signal production by relaying cells in the field), then the signal propagation periodicity will be similar to that obtained when there is one autonomous cell. If there were no stochastic variation in Ta , the most advanced cell would act as the pacemaker, the others being driven and not firing spontaneously, and T would equal Ta . If there were some noise, the cells which become most advanced in phase would become pacemakers and waves would propagate at some temporally varying minimum value ($T < Ta$). On the other hand if autonomous signal production is not entrainable (or the phase of the autonomous oscillator cannot be entrained to the refractory period of an autonomous cell) then a complex situation results. T becomes a function of the local values of Tr in the field and the randomly distributed firing of a population of pacemaker cells. With a few localized non-entrainable cells, regular beats will be propagated. With a large number, the continual autonomous firing will result in firing of the field at its refractory period ($T = Tr$).

A final possibility should be mentioned for pacemaker activity: this is that a local area of cells continuously secretes a level of transmitter substance (cAMP) which is above the relaying threshold of the field. There is reason to think that this kind of pacemaker activity may be important in late slime mould aggregation, in the functioning of the tip (Durstun, unpublished results; Drage, Robertson & Cohen, unpublished results). Such a pacemaker would propagate signals at the refractory period of the field.

6. The Origins of Spiral Wave Forms

The concentric wave forms which occur in early aggregation fields require only a point source pacemaker, e.g. an autonomous cell and sensitivity of the field of excitable cells for their existence. All of the other wave forms which have been described arise from interactions between waves and local inhomogeneities in aggregation fields. Fields of aggregating Dd cells are expected to be inhomogeneous in cell density and in their excitable properties. Some inhomogeneity will be random, and due to the initial distribution of the cells. Some will be nonrandom and contingent upon aggregation. Inhomogeneities will often break or distort wavefronts and, under appropriate conditions they may generate spirals. At least two types of inhomogeneity are important in this respect. Holes (inexcitable areas of aggregation fields) will contain a density of excitable cells which is below the critical percolation density for propagation of a wavefront. They may be areas which contain few cells.

Wiener & Rosenblueth (1946) have proposed a scheme whereby an interaction between waves and a hole may initiate a spiral. Their scheme was

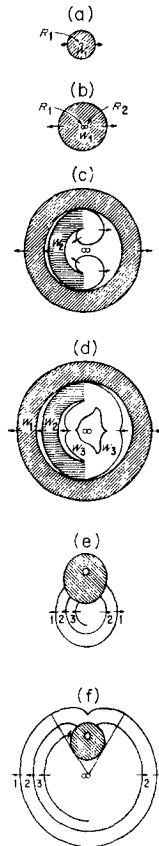


FIG. 2. Wiener & Rosenblueth's suggestion for generation of a spiral waveform via an interaction between wavefronts and a hole (a non-relaying area of an aggregation field).

(a) A point source R initiates a circular waveform (w_1).

(b) The medium is excited eccentrically near the center of the circular wave at R_2 .

(c) They propose that such eccentric excitation will make a new semi-circular wave (w_2) since medium on one side of the new point of excitation becomes sensitive before medium on the other side.

(d) Further evolution of the semi-circular waveform would make an extra quasicircular wave (w_3).

(e), (f) Interaction of wave w_1 , w_2 and w_3 with an appropriately sited hole could make a spiral.

devised to account for circulating waves of excitation (flutter) in the heart but it is applicable to slime mould aggregation waves. Their idea is that a semi-circular wavefront may be initiated by an appropriate interaction between a point source of stimulation and the refractory zone succeeding a wave. [Their figure is reproduced in Fig. 2(a), (b), (c), (d).] They use the semi-

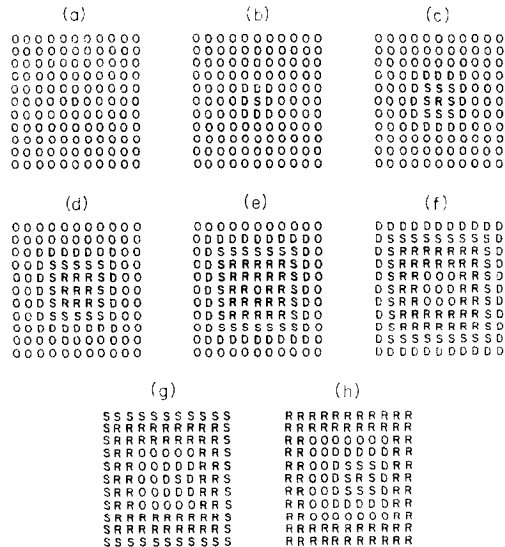


FIG. 3. Wiener & Rosenbluth's proposal is incorrect. Although excitation of R_2 (Fig. 2) may initially cause excitation of tissue to the left and not to the right of R_2 , the wave initiated by this excitation will be symmetrical (concentric). This is because the refractory zone succeeding wave w_1 (to the right of R_2) recedes rightwards at the same rate as the wave of excitation initiated at R spreads rightwards. This point is illustrated for a medium like a slime mould aggregation field, in which the rate of wave propagation is limited by a delay between sensing of a signal and signal relaying in each cell. The rules adopted are as follows:

- (1) Wave propagation is simplistically illustrated as being through a square lattice of cells.
- (2) Cells are initially sensitive to cAMP signals (0).
- (3) A cell sensing a signal enters a delay of one time unit before making a signal (cells marked D are in the delay state).
- (4) After one time unit, the cell makes an impulsive signal which instantaneously reaches all neighbours within one space unit (diagonal or orthogonal) of the lattice. A cell signalling is marked S for the following time unit.
- (5) For S and two time units after S (labeled R), the cell is refractory to further stimulation.
- (6) The cell then becomes sensitive again (0).
 - (a) $T = 0$. The sensitive lattice is stimulated centrally. The central cell adopts state D for one time unit.
 $T = 1$.
 - (b) D becomes S. A wave is initiated as the neighboring cells enter state D.
 - (c), (d), (e) $T = 2, 3, 4$.

The concentric wave propagates according to the rules set out above.

- (f) $T = 5$. The medium is stimulated eccentrically (right center of the lattice). The stimulated cell enters state D.
- (g) $T = 6$. D becomes S in the stimulated cell. The rightward cells leave state R and become sensitive simultaneously as the initiator cell leaves state D and makes a signal. Consequently the rightward cells now enter state D instead of state 0 and a concentric wave is initiated.
- (h) The wave stays concentric as it propagates.

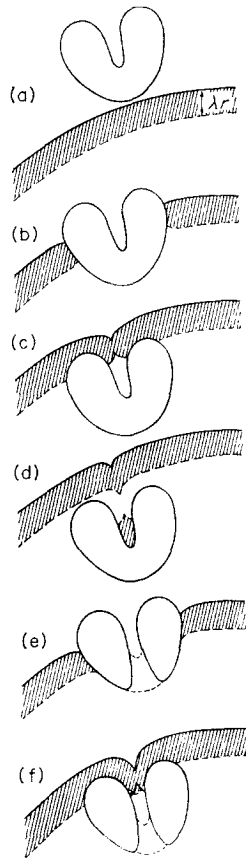


FIG. 4. Genesis of spirals by temporal change in a hole. (a) A wave approaches a cardioid hole. (b) The wavefront is broken by the hole. (c), (d) If the hole does not change, the outcome is an unbroken wavefront with a cusp. (e), (f) If the hole breaks into two, because of random movement of a sufficient number of sensitive cells into region *a*, two spirals can result.

circular wave thus generated to make a spiral, by interaction with an appropriate hole [as shown in their figure reproduced in Fig. 2(e), (f)]. As has been pointed out to me by Dr G. Mitchison and Professor M. H. Cohen, Wiener and Rosenblueth's proposal is incorrect. The fallacy in their argument becomes clear if we consider the effect of a point source of stimulation on a wave back [Fig. 3(a), (b), (c)]. The point source will make a circular wave and not a semi-circular wave, and a spiral cannot be made. It is possible however to use holes to make spirals in quite a trivial way, if we allow appropriate time dependent changes in an aggregation field. Random cell movements, for

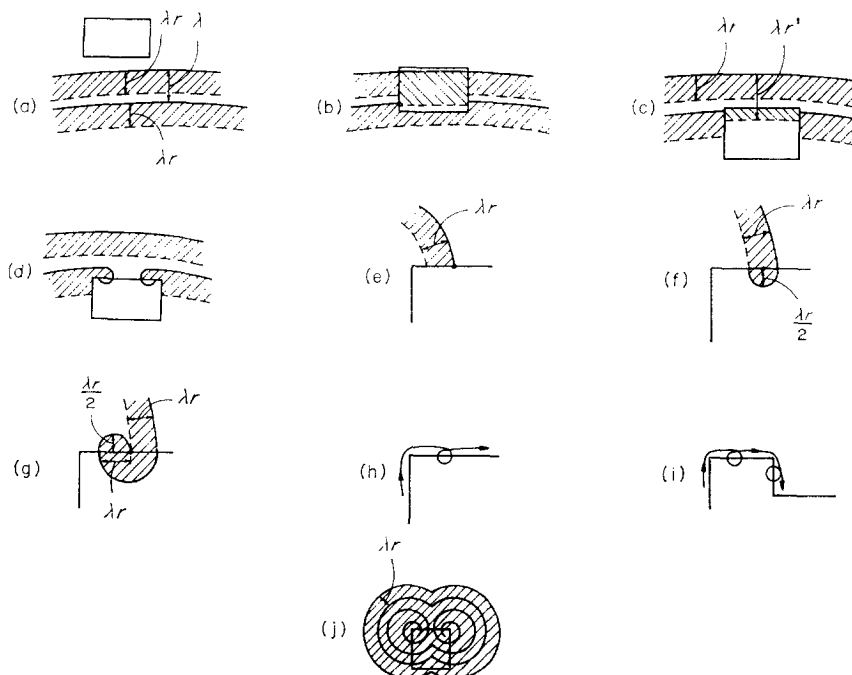


FIG. 5. An example of the initiation of a spiral via an interaction between two wavefronts and an area of non-uniform refractivity.

(a) Two wavefronts, separated by a wavelength (λ) which slightly exceeds the refractory wavelength of the medium (λr) approach a patch of tissue with a long refractory wavelength ($\lambda r'$) which exceeds λ . Shaded areas are refractory. Non-shaded areas are sensitive.

(b) The second wavefront is broken and develops two free ends, one at each side of the patch.

(c) The first wave leaves the patch, part of which remains refractory after the refractory zone of the first wave has left the surrounding tissue.

(d) The free ends of the second wave propagate around the patch until they reach a part of the patch which has become non-refractory and then spiral into it. There is a difficulty in making a stable spiral in this way, and this is as follows.

(e) Consider the point in time at which the patch of refractivity $\lambda r'$ becomes sensitive and allows entry of the external wavefront.

(f) The wavefront propagates semi-circularly through the patch for a radius of $\lambda r/2$ (until it reaches the back of the refractory zone of the externally propagating wave).

(g) It then re-enters the external medium and makes a second hemispherical external wave which cannot re-enter the $\lambda r'$ patch as long as the wave projectory is a straight line because it is perpetually λr behind the last propagating wavefront in the patch and therefore is within the refractory zone ($\lambda r'$) of this wave.

(h) The free end of the wavefront therefore makes only one turn and generates a transient spiral.

(i) A similar process will occur wherever the wavefront encounters a convex corner to the patch.

(j) Stable spirals may arise if the refractory period of the patch becomes equal to that of the field ($\lambda r' \rightarrow \lambda r$). If this occurs in the whole patch before completion of the first turn of the

example, can convert large holes into a number of smaller ones and some or all of the small holes may become foci for spirals if the cell movements coincide with an appropriate interaction between the hole and a wave, as shown in Fig. 4. There are other ways in which holes can initiate spirals. For example, Krinskii (1970) has shown that an inhomogeneity in refractory period, coupled with a hole, may start a spiral.

Hole spirals are expected to have a pacemaker loop which is the periphery of the hole and their period will be determined accordingly. They will undergo a defined evolution as described in section 7, and will finally become minimum spirals.

Inhomogeneities in the refractory period are anticipated in all Dd aggregation fields because the refractory period of Dd cells is an age dependent variable (about 7 min at the beginning of aggregation and about 2 min at the end of aggregation). Any stochastic variation in the developmental age of aggregating cells will therefore imply inhomogeneity in the refractory period of the aggregation field.

Patchiness in the refractory period of an aggregation field may lead to spiral initiation in the way shown in Fig. 5. Refractivity spirals (reverberators) have previously been discussed by Krinskii (1971) as a possible explanation for abnormal electrical activity (fibrillation) in heart muscle. They are expected to have the following characteristics. They will have the minimum wavelength and period possible for spirals in their local area of the aggregation field, as shown in Fig. 5. They may be transient (and may make only one wave) as shown in the figure. They may alternatively be stable, and can be stabilized in one or possibly two ways. Temporal changes in the refractory period of appropriate parts of the medium will sometimes occur and can make stable spirals, as shown in Fig. 5. Krinskii has also suggested that reverberators may have increased lifetimes if the duration of the excitatory signal (τ) is significantly long compared to the refractory period (Tr). The argument is given in detail in his paper (Krinskii, 1971). This kind of stabilization will not be important for Dd spirals if τ is short (0.1 to 0.2 sec) compared to Tr (100 to 460 sec) as proposed by Cohen and Robertson.

spiral a pair of counter-rotating spirals would be expected, as shown in the figure. Other outcomes are possible if only parts of the patch change their refractivity. The periods of refractivity spirals will be equal to the local refractory period of the external medium. This is a time dependent variable between 2 and 7 min, so their periods will be in this range and may change (decrease) with time.

Refractivity spirals might become stable or quasi-stable without any change in λr or $\lambda r'$ if the duration of the signal (T) were long relative to the refractory period (Tr), as discussed by Krinskii (1971). This kind of stability is not likely to be important for Dd refractivity spirals because (T) is thought to be very short (0.1 to 0.2 sec) compared to Tr (100 to 400 sec) in Dd (Drage, 1973).

7. Complications Due to the Movement Response

Dd cells propagating aggregation signals make a movement response which leads to their aggregation towards pacemaker regions. As was described earlier, the response is thought to be a unitary movement step, lasting about 100 sec and covering 20 μ , which appears to be elicited by an impulsive signal lasting 0.1 to 0.2 sec (Cohen & Robertson, 1971a). The nature of the signal has not yet been established. The published data indicate that it probably employs cAMP (Bonner *et al.*, 1969; Konijn, Barkley, Chang & Bonner, 1968), and that it is vectorial (since moving cells make an oriented response to centers). It is likely to be a transient gradient of cAMP. The movement response has some consequences for the properties of Dd aggregation fields as excitable media and the morphogenesis of Dd aggregation centers. These consequences and further properties of the movement response will now be described.

Cells aggregating to centers are attracted towards the nearest relaying cells. Therefore, azimuthal density inequalities in aggregation fields grow and aggregation fields separate into radial streams of cells (Cohen & Robertson, 1971b).

There is some evidence that aggregating Dd cells develop a fixed polarity for movement. When a cell makes a movement step, pseudopodia are extruded only from the front end of the elongated cell (Shaffer, 1957, 1962) and when a cell responds to successive signals from different orientations, it may sometimes turn, preserving its original polarity (G. Gerisch, personal communication). This fixed polarity of movement will contribute to the polarity of aggregation streams mentioned in section 3. Sections of streams fail to move smoothly towards aggregation centers if dissected out and replaced with reverse polarity (Bonner, 1950). Cells in aggregation streams have a location due to special cell junctions between the ends of cells, therefore a polar movement response in each cell would lead to streams having a fixed polarity for movement.

The movement response will lead to a definite course of evolution for spiral centers, engendered by interactions of waves with holes containing a sub-percolation density of relaying cells. The hole will first empty of cells as these are attracted to the signalling cells in its circumference and will then collapse, due to the centripetal movements of cells in the external aggregation field and will finally become a continuously excitable mass, fired by a stable minimum loop or other end product of spiral evolution.

8. Parameters of Dd Signal Propagation

The refractory period (Tr) has been measured as a function of time by watching the response of maturing fields of Dd cells to an artificial periodic

signal with a fixed period of 3 min. Preliminary results show that 6 hr cells respond to every third signal; 7 hr cells respond to every second signal and 9 hr cells respond to every signal. These data indicate that Tr is a time dependent variable with a range between > 6 and < 3 min (Drage, Robertson & Cohen, unpublished results). This fact could lead to interesting consequences for periods for wave propagation when aggregation fields are driven by autonomous cells. In particular in fields driven by one autonomous cell or many entrainable autonomous cells, if Ta is constant and Tr initially exceeds Ta or $m(Ta)$ and finally is less than Ta or $m(Ta)$, gating effects would be expected due to which the wave propagation period would decrease stepwise from $2Ta$ or $(m+1)(Ta)$ to Ta or $m(Ta)$. This prediction leads to the interpretation made of the periods of concentric Dd centers in paper 2 in this series. One other important variable which has been calculated is τ (the duration of the signal). τ is short (0.1 to 0.2 sec) compared to Tr (~150 to 400 sec) (Cohen & Robertson, 1971a, Drage, Robertson & Cohen, unpublished results). Therefore beats of the kind described by Krinskii (1970), due to interactions among τ , Ta and Tr are not anticipated in Dd aggregation fields.

I wish to thank Morrel Cohen and Anthony Robertson for their interest in and discussions of this paper.

This research was supported by N.I.H. Grant No. H4-04722 (General Medical Sciences) to the University of Chicago.

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